

Sep 17 Warmup

$$\int \frac{1}{x \sqrt{x+1}} dx$$

Need to use u-sub for this

$$u = \sqrt{x+1} \rightarrow u^2 - 1 = x$$

$$du = \frac{1}{2\sqrt{x+1}} dx \rightarrow du = \frac{1}{2u} dx \rightarrow dx = 2u du$$

$$\rightarrow \int \frac{1}{(u^2-1)u} \cdot 2u du = \int \frac{1}{(u^2-1)} 2 du = 2 \int \frac{1}{(u+1)(u-1)} du$$

Focusing just on this...

$$\frac{1}{(u+1)(u-1)} = \frac{A}{u-1} + \frac{B}{u+1}$$

$$1 = A(u+1) + B(u-1)$$

$$\text{If } u=1 \rightarrow 1 = A(2) + B(0)$$

$$\rightarrow 2A = 1 \rightarrow A = \frac{1}{2}$$

$$\text{If } u=-1 \rightarrow 1 = A(0) + B(-2)$$

$$\rightarrow -2B = 1 \rightarrow B = -\frac{1}{2}$$

$$2 \int \frac{A}{u-1} + \frac{B}{u+1} du$$

$$= 2 \int \frac{1/2}{u-1} - \frac{1/2}{u+1} du$$

$$= \int \frac{1}{u-1} du - \int \frac{1}{u+1} du$$

$$= \ln|u-1| - \ln|u+1| + C$$

$$= \ln|\sqrt{x+1}-1| - \ln|\sqrt{x+1}+1| + C$$

Subtracting natural logs is the same as one natural log where

thing inside is divided

$$= \ln \left| \frac{\sqrt{x+1}-1}{\sqrt{x+1}+1} \right| + C$$